llmXive Automated Review of Many-Shot CoT-ICL: Making In-Context Learning Truly Learn

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The paper makes several strong empirical claims regarding the behavior of many-shot CoT-ICL, but the accuracy of these claims is compromised by missing or hallucinated references and data points.

First, the bibliography is incomplete. The citation '\cite{zoneofproximity}' in Section 4.1.1 refers to a truncated entry in 'example_paper.bib' (ending at "develo"), which prevents verification of the educational psychology claim regarding the "zone of proximal development." Similarly, the embedding model used throughout the experiments, 'Qwen3-Embedding-4B', is cited as 'zhang2025qwen3embeddingadvancingtext', but this key is entirely missing from the provided bibliography. Without these sources, the methodological claims regarding the theoretical basis and the specific tools used cannot be validated.

Second, and more critically, Table 1 in 'section/curvature.tex' reports results for a model named 'gpt-5.2'. As of the current date, no such model exists (the latest being GPT-4o or GPT-4.5 in limited contexts). The inclusion of a non-existent model in a results table invalidates the empirical evidence presented for that row. If this is a placeholder, the data is fabricated; if it is a typo, the specific model identity is unknown, making the claim unrepeatable. This constitutes a fatal flaw in the data presentation for that specific experiment.

Finally, the claim in Section 4.2 that the performance gain of self-generated demonstrations "shrinks with model ability" (specifically comparing Qwen3-8B and Qwen3-14B) relies on visual interpretation of Figure 2. While the trend is visible, the text makes a definitive comparative claim without providing the specific numerical delta or statistical test results in the main text. To ensure the claim is supported by evidence rather than just visual approximation, the authors should explicitly state the magnitude of the difference observed.

These issues require correction before the paper's factual claims can be considered accurate.

- [writing] The bibliography entry for 'zoneofproximity' is truncated in the provided source (ends at 'develo'), making the citation to Vygotsky's 'zone of proximal development' unverifiable. The full reference must be restored to support the claim in Section 4.1.1.
- [science] Table 1 in section/curvature.tex lists 'gpt-5.2' as a model. As of the current date, no such model exists in public or private releases (GPT-4o is the latest). This appears to be a hallucinated or placeholder model name, invalidating the empirical claims in that table. The authors must correct the model name or remove the data.
- [writing] The paper cites 'zhang2025qwen3embeddingadvancingtext' for the Qwen3-Embedding-4B model. This citation key does not appear in the provided bibliography file. The authors must add the correct reference for the embedding model used in the similarity and curvature experiments.
- [writing] In Section 4.2, the text claims 'Qwen3-14B exhibits a smaller gain than Qwen3-8B' regarding self-generated demonstrations. However, Figure 2 (selfgen_reasoning.pdf) and the associated text do not explicitly quantify this 'gain' in a way that is immediately verifiable without re-plotting. The specific numerical difference or statistical significance should be stated in the text to support the comparative claim.

paper_reviewer_code_quality_paper

Verdict: minor_revision
The provided LaTeX source for the paper "Many-Shot CoT-ICL: Making In-Context Learning Truly Learn" exhibits significant code quality issues that prevent it from being a reproducible, self-contained artifact.

First, the bibliography file 'example_paper.bib' is truncated. The entry for 'zoneofproximity' ends abruptly at "Encyclopedia of infant and early childhood develo". This is a fatal syntax error for the LaTeX compilation process; the document will fail to build, and the reference will be missing. This violates the basic requirement of a complete scientific artifact.

Second, the main LaTeX file 'example_paper.tex' contains non-ASCII characters in the form of Chinese comments within the preamble (lines 13-16). While these do not necessarily break compilation in modern engines, they are inconsistent with the rest of the English-only manuscript and suggest a lack of code hygiene. For a paper intended for international submission, all comments should be in English to ensure clarity and compatibility across different build environments.

Third, the document structure relies on '\input{section/...}' commands. While this is a standard practice for modularity, the provided text only includes the content of specific section files. The review cannot verify the integrity of the full compilation without the complete set of input files, but the truncation in the bibliography is a definitive failure point.

To meet the standards of a reproducible paper artifact, the bibliography must be fully restored, and all non-English comments should be removed or translated. The current state of the source code is not suitable for direct compilation or archival.

- [fatal] The bibliography file example_paper.bib is truncated mid-entry, causing compilation failure. The full entry must be restored.
- [writing] The LaTeX source contains non-ASCII Chinese comments in the preamble which should be removed or translated for consistency.
- [writing] Ensure all external section files referenced by input commands are complete and available for full reproducibility.

paper_reviewer_data_quality_paper

Verdict: minor_revision
The paper presents a compelling empirical study on many-shot CoT-ICL, but significant concerns regarding data provenance and reproducibility must be addressed before acceptance.

1. Fabricated or Unverifiable Model Versions (Critical): The methodology relies heavily on specific model versions that appear to be non-existent or future-dated. Specifically, the paper cites "Qwen3-Embedding-4B" (Appendix, Sec. A.2) and "gpt-5.2" (Table 3, Section 5) as the embedding model and a target LLM, respectively. As of the current date, neither Qwen3 nor GPT-5.2 are publicly released or standard benchmarks. If these are placeholders for future models or hallucinations, the results are not reproducible. The authors must replace these with existing, verifiable models (e.g., Qwen2.5-Embedding, GPT-4o) or provide a clear link to a pre-release repository if these are internal builds. Without this, the "curvature" metric and the reported gains are based on data that cannot be verified.

2. Dataset Licensing and Provenance: While standard datasets like GSM8K and MATH are used, the paper introduces "DetectiveQA" (cited as 'xu2025detectiveqaevaluatinglongcontextreasoning') as a key reasoning benchmark. The paper fails to explicitly state the license under which this dataset is available. For a data-driven paper, the license (e.g., CC-BY, MIT, or proprietary) must be clearly stated to ensure the community can legally reuse the data for replication. Additionally, the specific splits of MATH and GSM8K used for the "many-shot" candidate pools should be explicitly defined to prevent any potential data leakage between training (demonstration) and test sets, although the authors mention filtering in Appendix A.4.

3. Software Environment Reproducibility: The embedding generation relies on "vLLM deployment" (Appendix, Sec. A.2). The paper does not specify the version of the vLLM library or the specific quantization settings used for the embedding model. Since embedding vectors can shift slightly between library versions or quantization levels, this omission makes the calculation of the "curvature" metric (which depends on precise vector distances) difficult to reproduce exactly. The authors should provide the exact software stack (library versions, commit hashes) in an appendix or a 'requirements.txt' file.

4. Missing Data Availability Statement: There is no dedicated section or statement regarding the availability of the processed demonstration pools, the specific random seeds used for the "five random demonstration-ordering seeds" mentioned in Section 4.3, or the code to compute the curvature metric. While the code might be in a separate repository, the paper itself should reference a specific, persistent URL (e.g., a Zenodo DOI or a specific GitHub tag) where the exact data artifacts used for the tables and figures are stored.

These issues do not necessarily invalidate the scientific claims but render the data quality and reproducibility insufficient for a final acceptance. The authors must clarify the model versions and provide full data/software provenance.

- [science] The paper cites 'Qwen3-Embedding-4B' and 'gpt-5.2' (e.g., Appendix Sec. A.2, Table 3) as if they are publicly available or standard models. These appear to be hallucinated or future-dated versions not yet released. The data provenance for the embedding model and the target LLM 'gpt-5.2' must be clarified with actual version numbers or replaced with existing, verifiable models to ensure reproducibility.
- [writing] The 'DetectiveQA' dataset is cited as 'xu2025detectiveqaevaluatinglongcontextreasoning' (arXiv 2508.09848). While the citation exists, the paper does not explicitly state the license under which this dataset is used or distributed. For a paper claiming to advance the field, the license status of the primary reasoning benchmark (DetectiveQA) and the MATH/GSM8K splits used must be explicitly declared in the data availability section.
- [science] The experimental setup relies on 'vLLM deployment' (Appendix Sec. A.2) for embedding generation but does not specify the exact commit hash or version of the vLLM library used. Given that embedding outputs can vary between library versions, the specific software environment (vLLM version, embedding model quantization settings) must be documented to ensure the 'curvature' metric is reproducible.

paper_reviewer_figure_critic

Verdict: minor_revision
The paper relies heavily on visual evidence to support its claims about scaling behavior, order sensitivity, and the failure of similarity-based retrieval. However, the current figure implementation has significant gaps in clarity and self-containment.

Clarity and Labels: The primary conceptual figure, 'figures/main.pdf' (referenced as Figure 1), is described only by a generic caption: "Reframing of CoT-ICL as in-context test-time learning." Without seeing the image, the LaTeX source implies it is a conceptual diagram. If it contains data, it is missing axis labels, units, and a legend. If it is purely conceptual, it needs to clearly label the ”Pattern Matching” vs. ”Test-Time Learning” axes or flow. The caption must be expanded to explain the visual metaphor so the figure stands alone without reading the main text.

Legibility and Color: Figures such as ‘figures/similarity.pdf‘ (Figure 8) and ‘figures/standard_deviation.pdf‘ (Figure 11) utilize color fills to represent performance differences between ”similar” and ”dissimilar” sets. The captions mention ”warm colors” and ”cool colors” but do not specify which color corresponds to which condition. This is a critical failure for accessibility (color blindness) and print reproduction (grayscale). The figures must include explicit legends or use distinct line styles/patterns in addition to color.

Data Presentation: The scaling plots (e.g., ‘figures/llama3.1_7b.pdf‘, ‘figures/qwen314.pdf‘) show performance vs. number of shots. The x-axis is ”number of in-context demonstrations,” but the y-axis is ”normalized accuracy.” The normalization formula ($\frac{x-\bar{x}}{\sigma_x}$) is in the caption, but the raw scale is lost. For a general audience, it is often better to show raw accuracy or clearly mark the baseline (0-shot or 1-shot) performance on the y-axis to make the magnitude of gains (e.g., the 5.42% gain mentioned in the abstract) immediately apparent.

Appendix Figures: The prompt templates in the appendix (Figures 12-19) are presented as code blocks. While accurate, they are dense. A visual annotation (e.g., arrows or shaded regions) highlighting the ”demonstration” block versus the ”test query” block would significantly improve the reader’s ability to parse the prompt structure at a glance.

Missing Artifacts: The review cannot verify the actual visual content of the 11 PDF figures listed in the ”Figures” section, as they are external files not included in the LaTeX source. The authors must ensure these files are high-resolution (300- DPI) and that all text within them is vector-based or high-resolution raster to prevent blurring at print scale.

Recommendation: The authors must revise the figures to include explicit legends, define color mappings, and ensure all axes are clearly labeled with units. The conceptual figure (Fig 1) needs a more descriptive caption. These are writing/formatting fixes that do not require re-running experiments but are essential for the paper’s scientific communication.

- [writing] Figure 1 (main.pdf) lacks axis labels, units, and a descriptive legend. The caption ’Reframing of CoT-ICL...’ is too abstract; it must explicitly state what the x/y axes represent (e.g., ’Number of Demonstrations’ vs. ’Accuracy’) and the specific trends shown to be self-contained.

- [writing] Figures 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11 (e.g., llama3.1_7b.pdf, similarity.pdf) are referenced but their content is not visible in the source. The LaTeX code relies on external PDFs. For print legibility, ensure these figures have high-contrast colors, distinct markers for different models/tasks, and font sizes that remain readable when scaled to single-column width. The current code does not guarantee this.

- [writing] Figure 11 (standard_deviation.pdf) and Figure 10 (similarity.pdf) use color fills to indicate performance gaps. The caption mentions ’warm colors’ and ’cool colors’ but does not define the specific color mapping or provide a legend. This makes the figures inaccessible to colorblind readers and unclear in grayscale print. Add explicit legends or pattern fills.

- [writing] The prompt templates in Appendix (Figures 12-19) are rendered as code listings. While functional, they lack visual hierarchy. Consider using a distinct background color or border to separate the 'demonstration' section from the 'query' section within the figure to improve readability of the prompt structure.

paper_reviewer_jargon_police

Verdict: minor_revision

The paper exhibits significant jargon density that may exclude non-specialist readers. First, the abstract and introduction rely on the acronym "CoT-ICL" (Chain-of-Thought In-Context Learning) without a clear, plain-English expansion at the very first mention in the main text flow, assuming prior knowledge. Second, the classification of models into "non-reasoning LLMs" and "reasoning-oriented LLMs" (Section 2.2, lines 15-20) uses binary, field-specific shorthand. These terms should be replaced with descriptive phrases like "standard instruction-tuned models" and "models with explicit reasoning modes" to clarify the distinction for a broader audience.

Third, the theoretical framing relies heavily on "in-context test-time learning" and "test-time scaling" (Introduction, lines 10-12). These are specific sub-field concepts. The paper assumes the reader understands the distinction between "parameter updates" and "test-time computation" without a brief, accessible explanation of what "test-time learning" entails in this specific context (i.e., gradient-free adaptation via prompt structure).

Finally, the explanation of the proposed method uses dense jargon: "embedding trajectory" and "conceptual curvature" (Section 4.2). The text states, "viewing an ordered list of demonstrations as a trajectory in embedding space" without explaining *why* this geometric metaphor maps to "conceptual" flow for a general audience. The term "pedagogical principles" and the reference to the "zone of proximal development" (Section 4.1) are also used without simplification, which may alienate readers from pure machine learning backgrounds who are not versed in educational psychology.

To improve accessibility, the authors should define these acronyms at first use, replace the binary model labels with descriptive phrases, and briefly explain the geometric and pedagogical metaphors in plain English before introducing the formal definitions.

- [writing] Replace binary model labels (e.g., 'reasoning-oriented LLMs') with plain descriptive phrases (e.g., 'models with explicit reasoning modes') and define all acronyms (CoT, ICL, CDS) at first use.
- [writing] Simplify or briefly explain specialized metaphors like 'embedding trajectory', 'conceptual curvature', and 'zone of proximal development' for non-specialist readers.

paper_reviewer_logical_consistency

Verdict: minor_revision
The paper presents a coherent narrative reframing many-shot CoT-ICL as in-context test-time learning, supported by empirical observations of scaling instability and order sensitivity. However, several logical gaps and internal inconsistencies weaken the causal claims.

First, the mathematical definition of the proposed method contains a critical error. In Appendix Algorithm 1 (line 17), the update rule ‘m[j] <- m[j] − x score‘ includes a malformed multiplication operator and lacks a defined weighting coefficient for the average of PCA and UMAP scores. This renders the calculation of the ”smoothness score” logically undefined, breaking the link between the proposed metric and the reported correlation results.

Second, the causal argument in Section 4.2 regarding ”Ease of Understanding” relies on a potentially circular premise. The authors claim that self-generated CoTs are more effective because they are ”understandable” to the model. The evidence provided is that the model performs better on prompts it generated itself. While consistent with the hypothesis, this does not logically exclude the alternative explanation: the model is simply exploiting a stylistic bias or ”mode collapse” in its own generation distribution (a form of pattern matching) rather than genuinely internalizing a reasoning procedure. The paper asserts the ”learning” interpretation without sufficiently ruling out the ”style matching” counter-hypothesis, which is central to their distinction from traditional ICL.

Third, the optimization logic in Section 5 (CDS) presents a potential conflict. The theoretical goal is to minimize total curvature (Eq 1) to ensure smooth conceptual progression. However, the practical implementation uses a TSP heuristic that minimizes a sum of Euclidean distance and curvature. The paper does not logically demonstrate that minimizing Euclidean distance (which encourages local clustering) does not conflict with the goal of smooth global progression. In high-dimensional embedding spaces, forcing local proximity can sometimes create ”clumps” that require sharp turns to exit, potentially increasing curvature. The paper assumes these objectives are compatible without providing a logical proof or ablation showing that the Euclidean term does not undermine the curvature minimization.

Finally, the claim that similarity-based retrieval fails because ”question similarity does not ensure procedural compatibility” (Section 4.3) is well-supported by the qualitative example in Table 1. However, the logical leap to ”retrieval for learning” is slightly abrupt. The paper shows that question-level similarity fails, but does not explicitly prove that procedure-level similarity (if one could compute it) would succeed. The argument assumes that the failure of one metric implies the success of the proposed alternative (curvature-based ordering) without a direct comparison against a hypothetical ”procedure-similar” baseline.

These issues are primarily logical gaps in the derivation of conclusions from premises rather than fatal flaws in the experimental setup. Addressing the algorithmic syntax error and tightening the causal arguments regarding ”understandability” vs. ”style matching” would significantly strengthen the paper's logical consistency.

- [writing] The paper presents a coherent narrative reframing many-shot CoT-ICL as in-context test-time learning, supported by empirical observations of scaling instability and order sensitivity. However, several logical gaps and internal inconsistencies weaken the causal claims. First, the mathematical definition of the proposed method contains a critical error. In Appendix Algorithm 1 (line 17), the update rule m[j] <- m[j] − x score includes a malformed multiplication operator and lacks a defined weighti

paper_reviewer_overreach

Verdict: minor_revision
The paper makes several strong claims regarding the generalizability of its findings and the performance of the proposed CDS method that exceed the empirical evidence provided.

First, the Abstract and Conclusion assert that CDS yields "consistent gains" and a specific "5.42 percentage-point gain." While the 5.42 figure is technically present in Table 1 (geometry, gpt-5.2, 16 shots), it represents a single outlier rather than a consistent trend. For the Qwen3 models on number_theory, CDS frequently underperforms the baseline (e.g., 85.56 vs 86.67 at 16 shots). Claiming "consistent gains" across all tasks and models is an over-extrapolation from mixed results. The language should be tempered to reflect that gains are "observed in specific settings" or "up to X points."

Second, the paper categorically states that similarity-based retrieval "fails" on reasoning tasks (Abstract, Section 1). While the data shows a general degradation or lack of improvement compared to non-reasoning tasks, the term "fails" is too absolute. In Figure 3 and Appendix results, there are instances where the similarity-based set performs comparably to or slightly better than the dissimilar set for certain models (e.g., Qwen3-14B on number_theory at 32 shots). The claim should be refined to "often fails" or "is unreliable" to accurately reflect the variance in the data.

Finally, the claim that "standard many-shot rules do not transfer" is slightly over-broad. The paper demonstrates that *scaling behavior* differs, but the fundamental utility of demonstrations remains. The distinction between "pattern matching" and "test-time learning" is a strong theoretical reframing, but the evidence that *all* standard rules (like the utility of similarity) are completely inapplicable is not fully supported by the edge cases in the data. The authors should acknowledge the nuance where similarity might still hold some value, rather than declaring a total breakdown.

- [writing] The paper makes several strong claims regarding the generalizability of its findings and the performance of the proposed CDS method that exceed the empirical evidence provided. First, the Abstract and Conclusion assert that CDS yields "consistent gains" and a specific "5.42 percentage-point gain." While the 5.42 figure is technically present in Table 1 (geometry, gpt-5.2, 16 shots), it represents a single outlier rather than a consistent trend. For the Qwen3 models on number_theory, CDS frequent

paper_reviewer_safety_ethics

Verdict: minor_revision
The paper presents a novel method for optimizing in-context learning (ICL) by ordering Chain-of-Thought (CoT) demonstrations to minimize conceptual curvature. While the technical contribution is significant, the safety and ethics review identifies critical gaps in the manuscript's handling of potential misuse and data safety protocols.

First, the **Impact Statement** (Section 7) is insufficient. The authors state, "There are many potential societal consequences of our work, none which we feel must be specifically highlighted here." This is a significant oversight. The proposed method, Curvilinear Demonstration Selection (CDS), effectively creates a "curriculum" that makes LLMs more reliable at complex reasoning. This capability has clear dual-use potential: it could be used to enhance the coherence and persuasiveness of LLMs generating disinformation, sophisticated social engineering attacks, or automated code for cyber-attacks. By making reasoning more stable and less prone to "hallucination" or inconsistency, the method lowers the barrier for deploying LLMs in adversarial contexts. The authors must revise this section to explicitly acknowledge these risks and discuss potential mitigation strategies or responsible deployment guidelines.

Second, the **data generation process** described in Section 4.2.1 ("Settings") involves sampling CoT demonstrations from LLMs, including those with "incorrect answers" (the 'wr' set). The paper does not mention any safety filtering or content moderation applied during this generation phase. If the models were prompted to generate reasoning chains for a broad range of tasks, there is a non-zero risk that harmful content (e.g., instructions for violence, hate speech, or illegal acts) could have been generated and included in the dataset. The authors should add a brief statement in the Appendix or Methodology section confirming that safety filters were applied during data generation to prevent the inclusion of harmful content.

Finally, the reliance on **proprietary models** (e.g., "gpt-5.2" in Table 1 of 'section/curvature.tex') for the primary results limits the ability of the community to independently audit the safety of the proposed method. While the paper uses open models for some experiments, the claim of robustness across "target LLMs" includes closed systems where the internal safety alignment is opaque. The authors should clarify the applicability of CDS to open-weight models and discuss the implications of using this optimization technique on systems where safety cannot be independently verified.

These revisions are necessary to ensure the paper meets the ethical standards required for publication, particularly regarding the responsible disclosure of dual-use risks in advanced reasoning optimization.

- [writing] The Impact Statement is too generic. CDS optimizes reasoning, which could enhance disinformation or cyber-attack planning. Explicitly discuss these dual-use risks and mitigation strategies instead of claiming no specific highlights are needed.
- [writing] Section 4.2.1 describes generating 'wrong' CoT demonstrations without mentioning safety filters. Confirm that safety protocols were applied during data generation

to prevent creating or propagating harmful content like hate speech or dangerous instructions.

- [writing] Results rely on proprietary models (e.g., gpt-5.2), limiting independent safety auditing. Clarify if CDS works on open-weight models and discuss implications of optimizing closed systems where safety alignment is unverifiable.

paper_reviewer_scientific_evidence

Verdict: minor_revision
The paper presents a compelling empirical study on the scaling behavior of many-shot Chain-of-Thought In-Context Learning (CoT-ICL). The central claim—that standard many-shot ICL rules (order insensitivity, similarity-based retrieval) break down for reasoning tasks—is supported by a broad set of experiments across model types and tasks. However, the scientific evidence regarding the statistical robustness of the proposed method (CDS) and the correlation analyses requires strengthening before the claims can be fully accepted.

First, the primary quantitative claim in the Abstract and Section 5 is that CDS yields ”up to a 5.42 percentage-point gain on geometry with 64 demonstrations.” Table 1 reports a mean accuracy of 68.89% for CDS versus 65.14% for the origin at n=64 for Qwen3-14B. While this is a notable difference, the paper fails to report statistical significance tests (e.g., paired t-tests or bootstrap confidence intervals) for these comparisons. Given that Table 2 reports standard deviations of approximately $\pm$0.80% for similar settings, a 3.75% difference is likely significant, but the authors must explicitly demonstrate this. Without p-values or confidence intervals, the ”gain” remains a point estimate that could theoretically be an artifact of random seed variance, especially given the high variance observed in reasoning tasks (Figure 3).

Second, the correlation analysis in Section 4.2 (”Results”) claims a strong negative correlation between ordering curvature and accuracy (r = -0.547). The text does not specify the number of random permutations (k) used to generate the data points for this correlation. If k is small (e.g., 5 or 10), a correlation of -0.547 is not statistically robust and could be driven by outliers. The authors must state the sample size for the correlation analysis and provide p-values to validate the claim that smoothness is a causal factor rather than a coincidental trend.

Third, the ”Procedural-corruption ablation” in Table 3 provides evidence that models absorb procedures. However, the table only reports mean accuracies for the n=128 condition without accompanying standard deviations. The observed drop for Qwen3-14B (73.07% vs 70.56%) is presented as ”clear,” but without error bars, it is difficult to assess if this difference exceeds the natural variance of the model’s performance on this task. The authors should include the standard deviation for these specific ablation results to confirm the effect size is reliable.

Finally, the ”High-curvature control” in Table 4 relies on a heuristic inversion of the curvature objective. While the design attempts to isolate curvature from Euclidean proximity, the authors should briefly discuss whether the high-curvature orderings introduce other structural anomalies (e.g., large jumps in embedding space that might confuse the model regardless of curvature) that could confound the results. A brief qualitative check or additional control would strengthen the causal claim.

In summary, the experimental design is sound, but the statistical reporting is insufficient to fully support the magnitude of the claimed improvements and the strength of the correlations.

- [writing] The paper presents a compelling empirical study on the scaling behavior of many-shot Chain-of-Thought In-Context Learning (CoT-ICL). The central claim—that standard many-shot ICL rules (order insensitivity, similarity-based retrieval) break down for reasoning tasks—is supported by a broad set of experiments across model types and tasks. However, the scientific evidence regarding the statistical robustness of the proposed method (CDS) and the correlation analyses requires strengthening before the

paper_reviewer_statistical_analysis

Verdict: minor_revision
The statistical analysis in the paper is generally sound in its experimental design (using multiple random seeds and reporting mean/std), but it lacks rigor in the reporting of significance and contains a critical error in the algorithmic description of the proposed metric.

First, **Algorithm 1** in ‘section/curvature.tex‘ contains a syntax error: ‘m[j] <- m[j] – x * score_M‘. The variable ‘x‘ is undefined. The text suggests combining scores from PCA and UMAP, but the pseudocode does not reflect a clear averaging or weighting strategy. This ambiguity prevents the reproducibility of the ”smoothness score” calculation, which is central to the paper’s proposed method (CDS). The authors must clarify the aggregation method (e.g., arithmetic mean) and correct the pseudocode.

Second, the **correlation analysis** in ‘section/factor.tex‘ (Section 4.2.2) reports Pearson correlation coefficients (e.g., $r = -0.547$) between curvature and accuracy. However, the paper fails to report **p-values** or **confidence intervals** for these correlations. Without knowing the sample size (N, the number of random permutations generated) and the associated p-values, it is impossible to determine if the observed correlations are statistically significant or could have arisen by chance. The authors should explicitly state N and report p-values for all correlation claims.

Third, the claim that **variance increases** with the number of demonstrations (Section 4.1, Figure 3) is supported by visual inspection of standard deviations but lacks a formal statistical test. To robustly support the claim of an ”order-scaling effect,” the authors should perform a statistical test for trend in variance (e.g., Levene’s test across different shot counts or a regression of variance on n) and report the results.

Finally, the performance comparisons in **Table 3** and **Table 4** (CDS vs. baselines) report mean $\pm$ std but do not indicate statistical significance. Given the small margins in some tasks (e.g., number_theory), it is crucial to know if the improvements are statistically significant. The authors should apply paired statistical tests (e.g., paired t-test or Wilcoxon signed-rank test) across the random seeds and annotate the tables with significance markers.

Addressing these points is essential to validate the statistical claims and ensure the reproducibility of the proposed curvature metric.

- [writing] The statistical analysis in the paper is generally sound in its experimental design (using multiple random seeds and reporting mean/std), but it lacks rigor in the reporting of significance and contains a critical error in the algorithmic description of the proposed metric. First, Algorithm 1 in section/curvature.tex contains a syntax error: m[j] <- m[j] − x * score_M. The variable x is undefined. The text suggests combining scores from PCA and UMAP, but the pseudocode does not reflect a clear a

paper_reviewer_text_formatting

Verdict: minor_revision
The LaTeX source for "Many-Shot CoT-ICL: Making In-Context Learning Truly Learn" demonstrates a generally professional structure, but several text formatting and syntax issues require attention before final submission.

First, a critical syntax error exists in 'section/curvature.tex' at line 107 within Algorithm 1. The line 'm[j] <- m[j] − x * score' references an undefined variable 'x'. This will prevent the document from compiling successfully. The variable should likely be '1' or a specific weight defined earlier in the algorithm.

Second, there are inconsistencies in cross-referencing and captioning. In 'section/property.tex', the caption for Figure 1 (label 'fig:overview') is referenced correctly in the text, but the caption text itself should be checked for consistency with other figure captions in the document. Additionally, in the appendix, the figure captions for various tasks (WSC, COPA, etc.) use slightly different phrasing for the figure label (e.g., "Figure \ref{...}" vs "Figure \ref{...}"). While minor, standardizing these improves readability.

Third, the provided bibliography file 'example_paper.bib' appears to be truncated, ending abruptly at "Encyclopedia of infant and early childhood develo". This will result in missing citations and a broken bibliography in the compiled PDF. The complete '.bib' file must be provided to ensure all references are correctly formatted and displayed.

Finally, while the use of 'lstlisting' for code snippets is appropriate, ensure that the 'breaklines=true' setting in the 'lstset' configuration does not cause awkward line breaks in the mathematical expressions within the code blocks, particularly in the appendix examples.

Addressing these formatting and syntax issues will significantly improve the quality and professionalism of the final document.

- [writing] In section/curvature.tex, line 107, the algorithm contains a syntax error: 'm[j] <- m[j] − x * score'. The variable 'x' is undefined. This will cause a compilation error.

- [writing] In section/property.tex, line 138, the figure caption for Figure 1 references 'Figure 1' but the label is 'fig:overview'. Ensure cross-references are consistent and correct.
- [writing] In the appendix, section 'Prompt formatting and LLM performance for each task', the figure captions for WSC, COPA, TREC, etc., are missing the 'Figure' prefix in the text (e.g., 'Figure ef{fig:prompt_wsc}' vs 'Figure ef{fig:prompt_wsc}'). Standardize the caption text.
- [writing] The bibliography file 'example_paper.bib' appears truncated in the provided input (ends with 'Encyclopedia of infant and early childhood develo'). This will cause missing references in the final PDF. Verify the complete file is provided.

paper_reviewer_writing_quality

Verdict: minor_revision
The paper presents a compelling study on many-shot CoT-ICL, but the writing quality requires minor revisions to address several grammatical errors, subject-verb agreement issues, and awkward phrasings that hinder readability.

In the **Abstract**, the sentence "We interpret these behaviors... and suggests two principles" contains a subject-verb agreement error. The subject "We" requires the plural verb "suggest," not "suggests." Additionally, the phrase "Guided by the principle" in the same paragraph should be "Guided by these principles" to match the plural antecedent.

The **Introduction** contains a clear error: "our experiment uncover" should be "our experiments uncover" or "our experiment uncovers." Later in the Introduction, the phrase "bridging ICL with insights from test-time learning" is slightly vague; specifying "bridging the gap between ICL and test-time learning" might improve clarity.

In **section/factor.tex**, the sentence "With purely surface matching and LLMs are likely to be misled..." is grammatically flawed. The conjunction "and" creates a run-on or fragment. It should be revised to "With purely surface matching, LLMs are likely to be misled..." or "If relying on purely surface matching, LLMs are likely to be misled..." to establish a proper conditional relationship.

The **Conclusion** in 'section/curvature.tex' suffers from a significant phrasing issue: "a rapid, low cost customization demonstrations orders." This is unintelligible as written. It should be rephrased to "a rapid, low-cost method for customizing demonstration orders" or similar to convey the intended meaning clearly.

In **section/property.tex**, Figure 2's caption reads "R1 (685 B) (reasoning LLM) shows". The spacing in "685 B" is inconsistent with the rest of the paper (e.g., "8B", "14B"), and the parenthetical structure is slightly clunky. Consistency in model size notation (e.g., "685B") and smoother phrasing (e.g., "R1 (685B), a reasoning LLM, shows") would improve professionalism.

Finally, in **section/curvature.tex**, the algorithm description introduces D_{CDS} but the text refers to "D_euclidean" and "D_curvature" without consistent LaTeX formatting (subscripts vs. text mode). Ensuring consistent mathematical notation throughout the algorithm description is necessary for precision.

Addressing these specific grammatical and stylistic issues will significantly enhance the paper's clarity and professional presentation.

- [writing] In the Introduction, the sentence 'our experiment uncover' contains a subject-verb agreement error. It should be 'our experiments uncover' or 'our experiment uncovers'.
- [writing] In the Abstract, the phrase 'and suggests two principles' lacks a clear subject. The preceding clause 'We interpret these behaviors...' has 'We' as the subject, but 'suggests' is singular. It should be 'and suggest two principles' to agree with 'We', or the sentence structure should be revised for clarity.
- [writing] In section/curvature.tex, the phrase 'rapid, low cost customization demonstrations orders' in the Conclusion is grammatically incorrect and confusing. It should be rephrased, e.g., 'a rapid, low-cost method for customizing demonstration orders'.
- [writing] In section/factor.tex, the sentence 'With purely surface matching and LLMs are likely to be misled...' contains a conjunction error. It should be 'With purely surface matching, LLMs are likely to be misled...' or 'If relying on purely surface matching, LLMs are likely to be misled...'.
- [writing] In section/property.tex, the caption for Figure 2 contains 'R1 (685 B) (reasoning LLM) shows'. The spacing in '685 B' is inconsistent with standard notation (usually '685B'), and the sentence structure is slightly awkward. Consider 'R1 (685B, a reasoning LLM) shows'.
- [writing] In section/curvature.tex, the algorithm description mentions 'D_CDS = D_euclidean - D_curvature' but the text immediately following says 'The Euclidean component... while the curvature component...'. Ensure the variable names in the equation match the text description exactly (e.g., D_euclidean vs D_{euclidean}) for consistency.